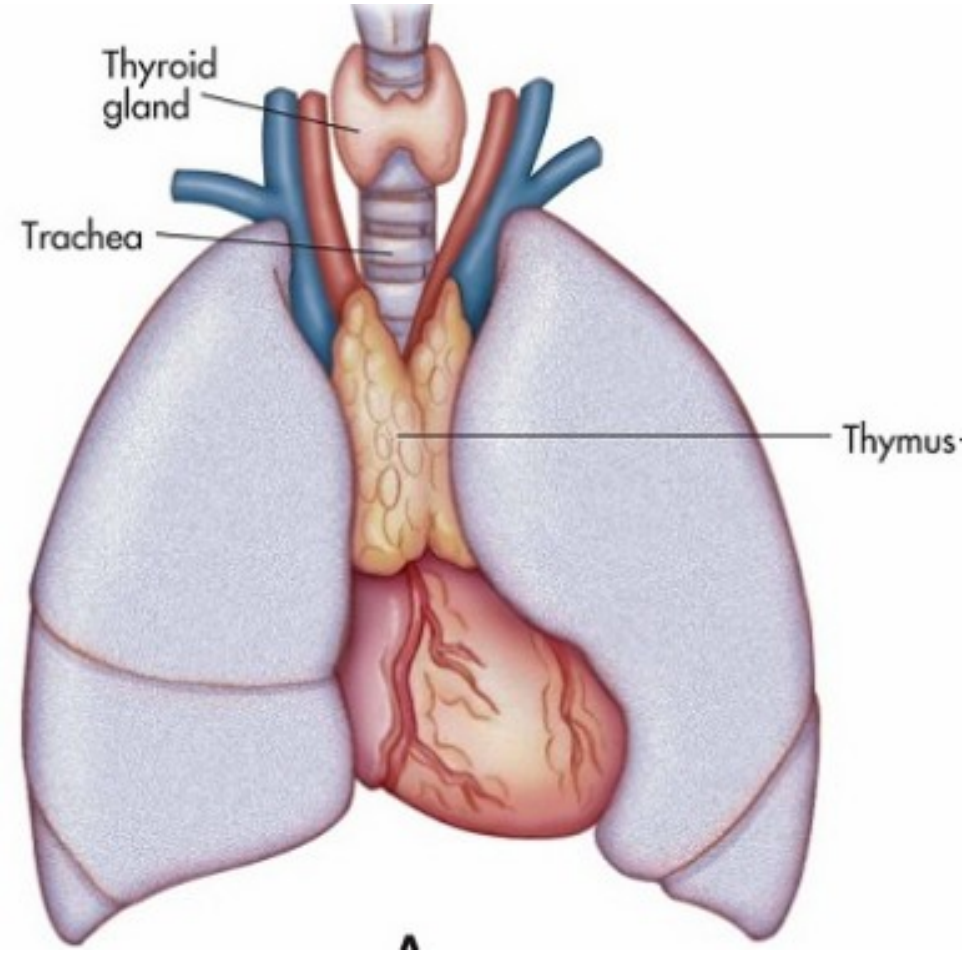
Other Endocrine Organs and Tissues

- Thymus
- Pineal
- Gonads
- And more…



The Thymus

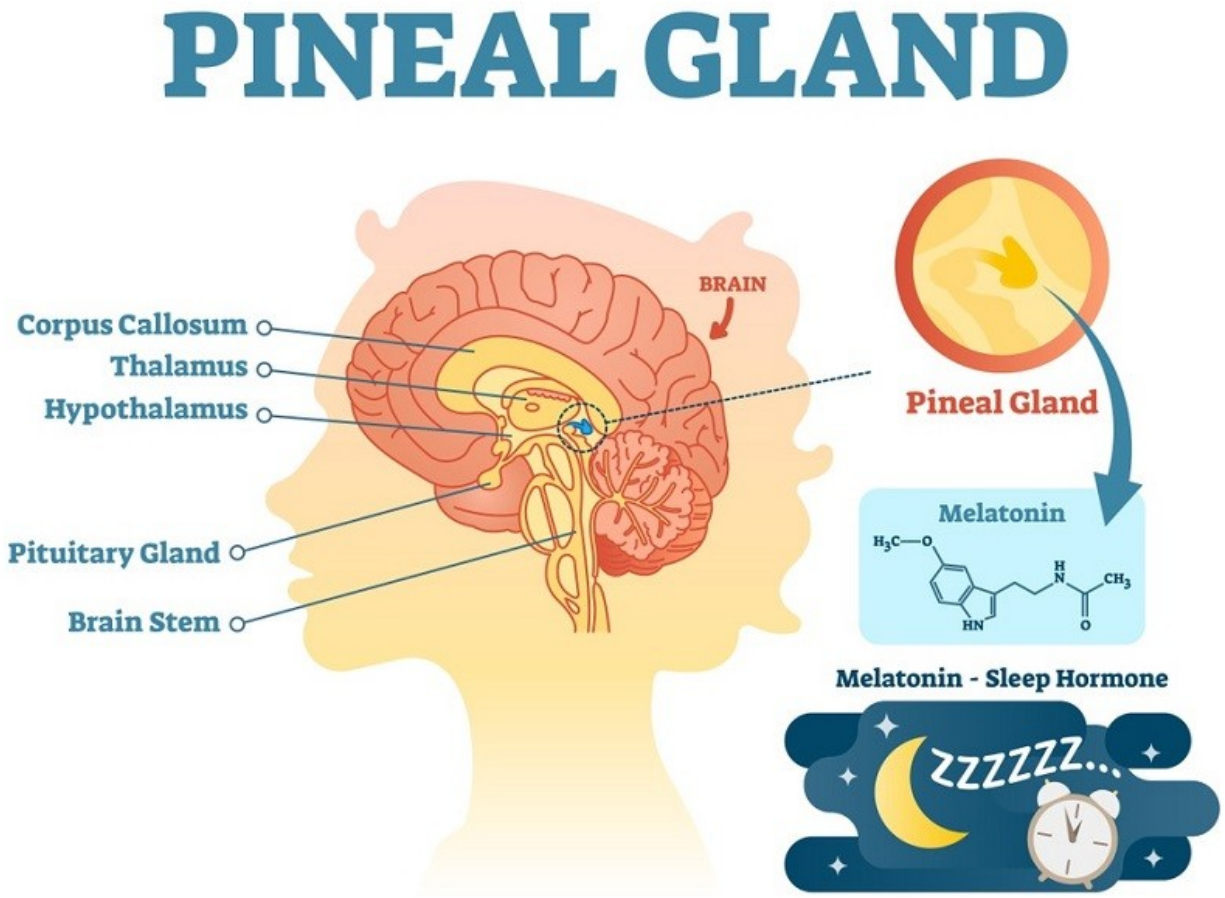
- Largest in infants and children, shrinks in adulthood
- Located in top of chest
- Secretes Thymosins
- Thymosins help mature T-lymphocytes – involved in immunity



Learning Aim: Describe the location and function of the thymus

The Pineal Gland

- Found deep in the brain
- Size of pea in childhood, decreases in size and eventually calcifies during adulthood and old age.
- Secretes *melatonin* – involved in regulating sleep patterns.
    - Secretion of melatonin is stimulated by darkness and inhibited by light



Learning Aim: Describe the location and function of the pineal

The Gonads

- Ovaries and Testes
  - Stimulated by FSH and LH from the anterior pituitary after signalling with GnRF from hypothalamus
  - Release hormones:
    - *Oestrogen* and *Progesterone*: maintain menstrual cycle, development and maintenance of female sex characteristics.
    - *Androgens (eg Testosterone):* male sex hormones. Develop and maintain male sex characteristics.



Note: These are not in the syllabus for Year 12 Human Biology so should not be asked in any assessment.

Learning Aim: Describe the function of the gonads and the hormones they produce.

Other endocrine tissues

- Stomach and small intestine:
  - Secrete some hormones that coordinate digestion
- Kidneys:
  - Secrete *erythropoietin* (EPO) – stimulates RBC production.
- Heart
  - Secretes *atrionatriuretic factor* – helps reduce blood pressure – interacts with aldosterone regulation
- Placenta
  - Secretes hormones that help to maintain pregnancy, assist foetal development and stimulate lactation.

Learning Aim: Be aware of other organs and tissues that secrete hormones